# 山东大学 计算机科学与技术 学院

# 计算机组成与设计 课程实验报告

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|---|---|---|
| 实验题目：实验 13 控制器实验 | | |
| 实验学时：2 | 实验日期： 2023.12.22 | |

**实验目的：**

目前控制器设计大都采用微程序设计方法，又称存储逻辑控制器。微程序控制器电路结构如图 13-1 所示。它由控制存储器 CROM、微程序 PC 计数器和微指令寄存器 IR 构成。

**实验软件和硬件环境：**

**软件环境：**

QuartusII 软件

**硬件环境：**

1. 实验室台式机
2. 计算机组成与设计实验箱

**实验原理和方法：**

其中，微程序计数 PC 向控制存储器提供 8 位微地址，在控存读信号 的作用下，读出一条长 24 位的微指令代码，并在打入命令 CP IR 的作用下，送入 IR。

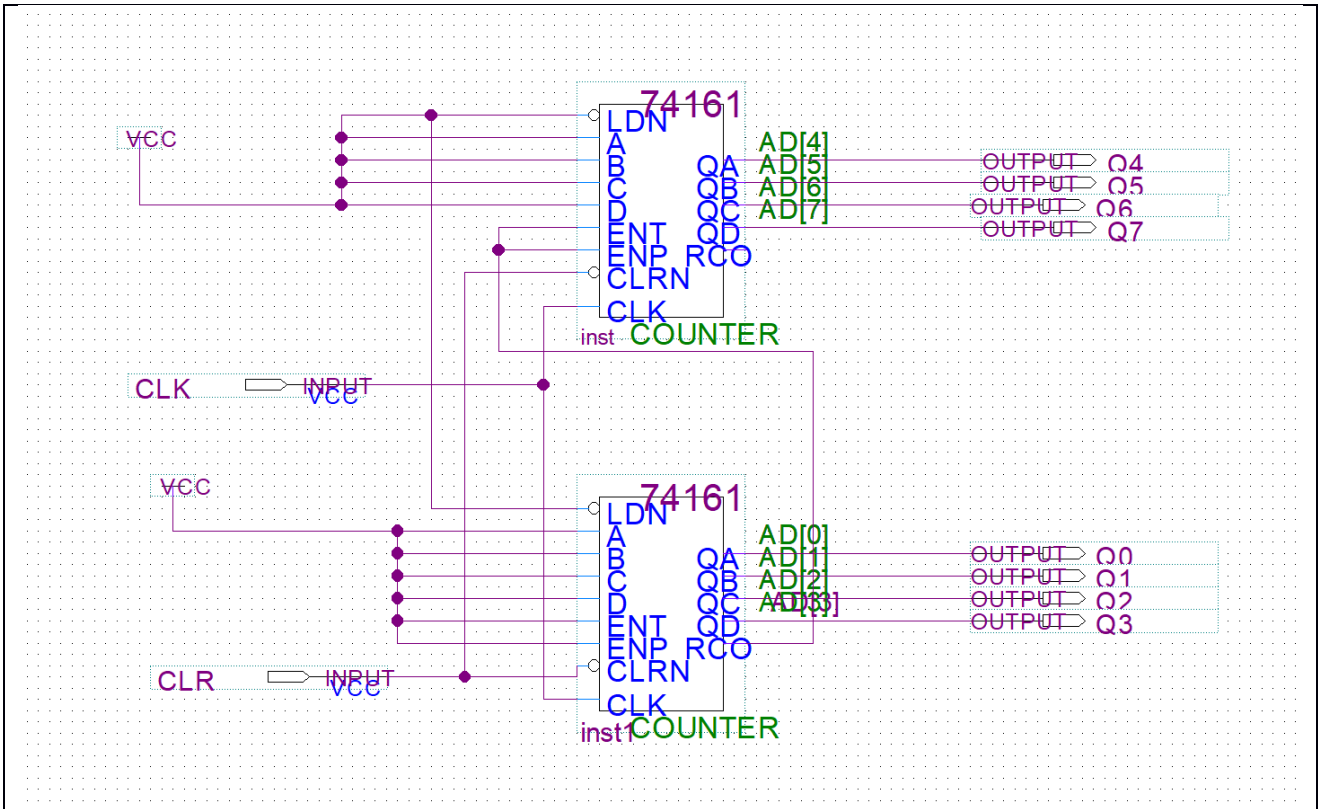
每当按一次脉冲键便产生一个负脉冲，该脉冲的作用是：

- 作为读控存的命令 。
- 负脉冲当作 CP IR 将读出的微指令打入微指令寄存器 IR。
- 负脉冲的上升沿使 PC＋1 形成下一条微指令的地址。
- 负脉冲反相后的上升沿作为寄存器打入脉冲。

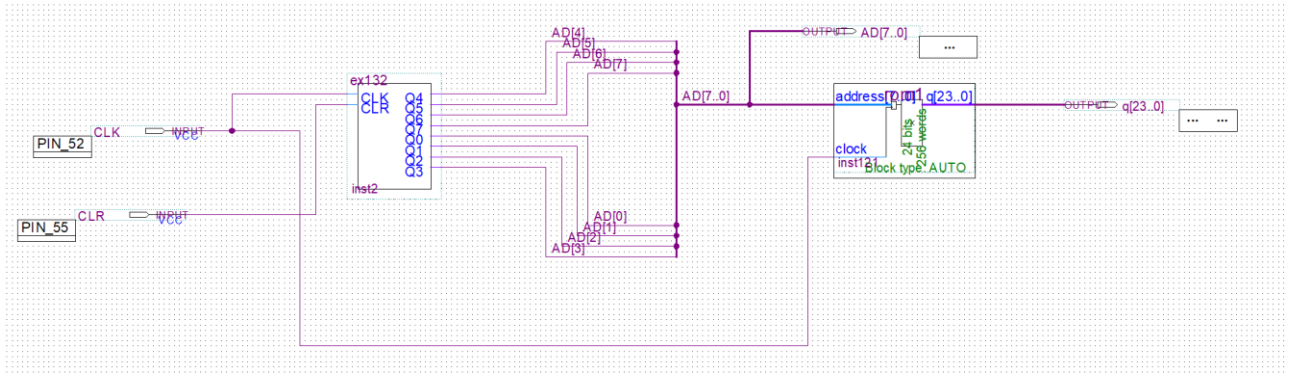
实验步骤：

（1） 原理图输入：根据原理图，完成电路图的连接，电路图连接如下。

首先完成计数器的设计，并生成计数器元件（下图为计数器）：




其次，将计数器元件与 ROM 芯片连接，电路图连接如下：




（2） 管脚锁定：引脚连接如下图：

| Node Name | Direction | Location | I/O Bank | REF Grou | ter Locati | O Standa | Reserved | ent Strer | Slew Rate | erential I |
|---|---|---|---|---|---|---|---|---|---|---|
| AD[7] | Output | PIN_39 | 3 | B3_N0 | PIN_39 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| AD[6] | Output | PIN_42 | 3 | B3_N0 | PIN_42 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| AD[5] | Output | PIN_83 | 5 | B5_N0 | PIN_83 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| AD[4] | Output | PIN_77 | 5 | B5_N0 | PIN_77 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| AD[3] | Output | PIN_74 | 5 | B5_N0 | PIN_74 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| AD[2] | Output | PIN_70 | 4 | B4_N0 | PIN_70 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| AD[1] | Output | PIN_65 | 4 | B4_N0 | PIN_65 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| AD[0] | Output | PIN_60 | 4 | B4_N0 | PIN_60 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| CLK | Input | PIN_52 | 3 | B3_N0 | PIN_52 | 2.5 ...ult) | | 8mA...lt) | | |
| CLR | Input | PIN_55 | 4 | B4_N0 | PIN_55 | 2.5 ...ult) | | 8mA...lt) | | |
| q[23] | Output | PIN_144 | 8 | B8_N0 | PIN_144 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[22] | Output | PIN_143 | 8 | B8_N0 | PIN_143 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[21] | Output | PIN_142 | 8 | B8_N0 | PIN_142 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[20] | Output | PIN_141 | 8 | B8_N0 | PIN_141 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[19] | Output | PIN_138 | 8 | B8_N0 | PIN_138 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[18] | Output | PIN_137 | 8 | B8_N0 | PIN_137 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[17] | Output | PIN_136 | 8 | B8_N0 | PIN_136 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[16] | Output | PIN_135 | 8 | B8_N0 | PIN_135 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[15] | Output | PIN_125 | 7 | B7_N0 | PIN_125 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[14] | Output | PIN_128 | 8 | B8_N0 | PIN_128 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[13] | Output | PIN_114 | 7 | B7_N0 | PIN_114 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[12] | Output | PIN_120 | 7 | B7_N0 | PIN_120 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[11] | Output | PIN_105 | 6 | B6_N0 | PIN_105 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[10] | Output | PIN_113 | 7 | B7_N0 | PIN_113 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[9] | Output | PIN_100 | 6 | B6_N0 | PIN_100 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[8] | Output | PIN_101 | 6 | B6_N0 | PIN_101 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[7] | Output | PIN_80 | 5 | B5_N0 | PIN_80 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[6] | Output | PIN_85 | 5 | B5_N0 | PIN_85 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[5] | Output | PIN_73 | 5 | B5_N0 | PIN_73 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[4] | Output | PIN_76 | 5 | B5_N0 | PIN_76 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[3] | Output | PIN_71 | 4 | B4_N0 | PIN_71 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[2] | Output | PIN_72 | 4 | B4_N0 | PIN_72 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[1] | Output | PIN_68 | 4 | B4_N0 | PIN_68 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| q[0] | Output | PIN_69 | 4 | B4_N0 | PIN_69 | 2.5 ...ult) | | 8mA...lt) | 2 (d...ult) | |
| <<ne...de>> | | | | | | | | | | |

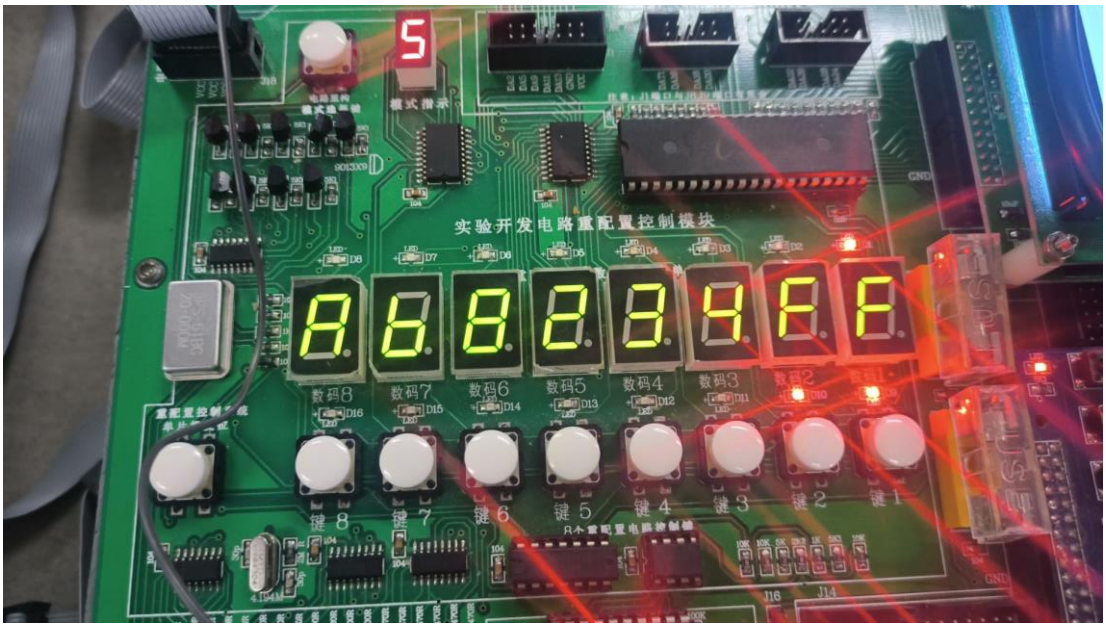
（3） rom 芯片中每个存储单元内容设置如下：

| Index | Instance | Status | Width | Depth | Type | Mode |
|---|---|---|---|---|---|---|
| 0 | NONE | Unexpe... | 24 | 256 | RAM/ROM | Read/Write |

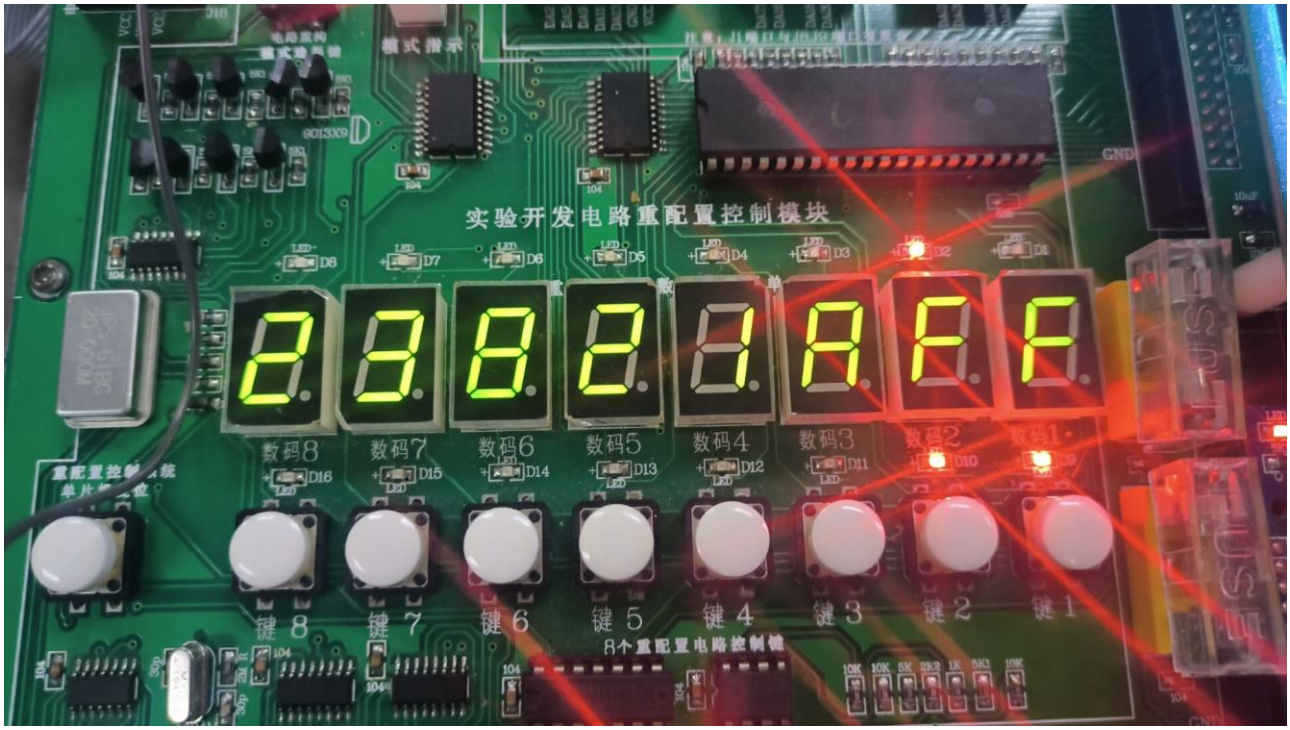
Instance 0: NONE

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 000000 | 91 21 12 | AB 82 34 | 23 82 1A | 91 12 22 | 23 20 01 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 00000c | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 000018 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 000024 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 000030 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 00003c | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 000048 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 000054 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 000060 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 00006c | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 000078 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 000084 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 000090 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 00009c | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 0000a8 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 0000b4 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 0000c0 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 0000cc | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 0000d8 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 0000e4 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 0000f0 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | . |
| 0000fc | 00 00 00 | 00 00 00 | 00 00 00 | 00 00 00 | | | | | | | | | | | | . |

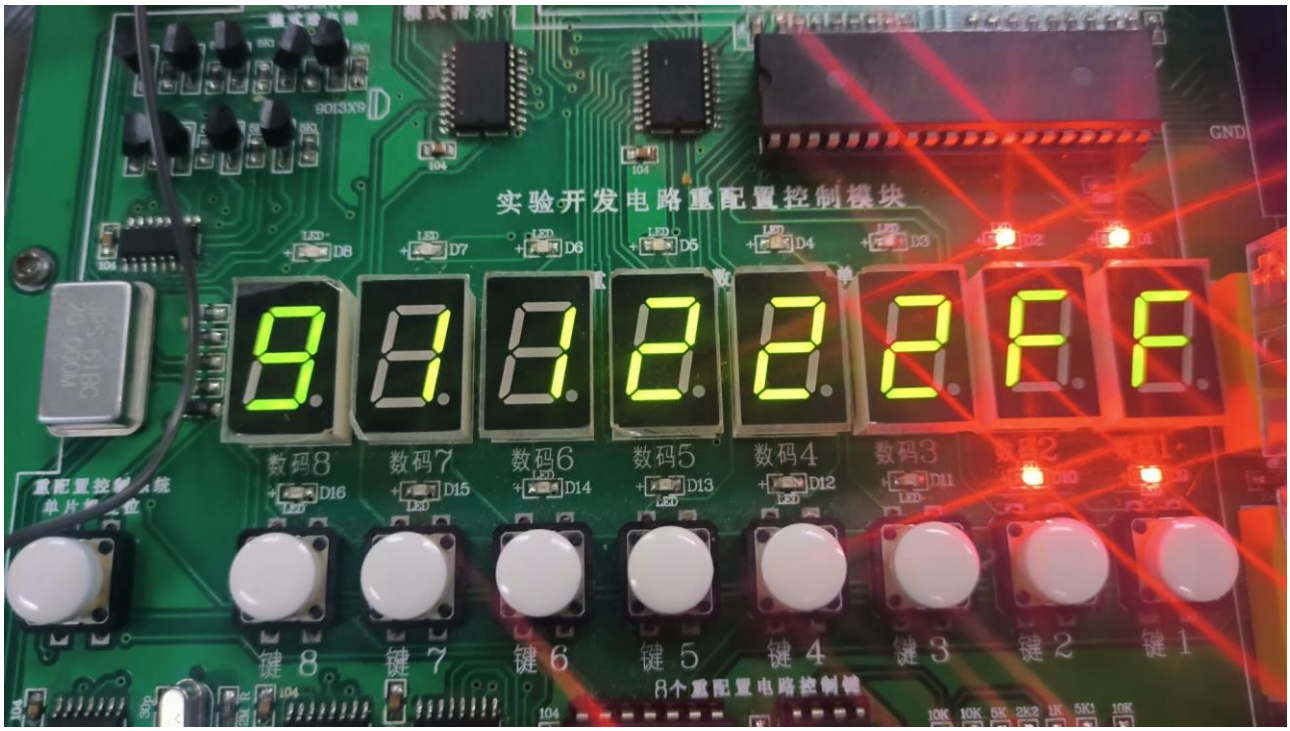
首先，发出一次时钟信号，将地址 1 中的数据读出，数据结果如下图：




然后，再发出第二次时钟信号，将地址 2 中的数据读出，结果读出如下图：



最后，再发出第三次时钟信号，将地址 3 中的数据读出，结果读出如下图：




**结论分析与体会：**

这次的实验让我对控制器有了一个较深入的理解，学会了如何利用人工设计计数器和 ROM 芯片来完成芯片某一特定地址存储单元的内容读出。